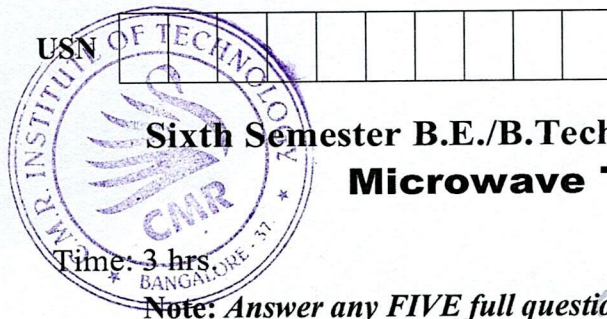




Max. Marks: 100

Module-1

- 1 a. Explain the operating principle of 9 Gunn-decode with neat energy band diagram and I-V-Characteristics. (08 Marks)
- b. A Telephone Line has $R = 6 \Omega/\text{km}$, $L = 2.2 \text{ mH/km}$, $C = 0.005 \text{ mF/km}$ and $G = 0.05 \text{ mho/km}$. Determine Z_0 , α , β at 1 KHz frequency and also find phase – velocity V_p (m=milli). (08 Marks)
- c. What are standing waves? Obtain expressions for VSWR in terms of reflection coefficient(P). (04 Marks)

OR

- 2 a. Derive the transmission line equations in voltage and current forms. (08 Marks)
- b. A 50Ω lossless line connects a matched signal of 100 KHz to a load of 100Ω . The load power is 100 mW. Calculate :
i) Voltage reflection coefficient (P)
ii) VSWR of the load
iii) Position of first $V_{\min}$ and $V_{\max}$
iv) Imp at $V_{\min}$ and $V_{\max}$ and value of $V_{\min}$ and $V_{\max}$. (08 Marks)
- c. Explain briefly single stub impedance matching technique. (04 Marks)

Module-2

- 3 a. Derive the S-matrix representation of a multiport network. (08 Marks)
- b. What is circulator? Explain the operating principle of 4 port circulator. (06 Marks)
- c. What is attenuators? Explain its different types briefly. (06 Marks)

OR

- 4 a. Derive the S-matrix for Magic –T. (08 Marks)
- b. What are waveguide Tees? Explain briefly each type. (08 Marks)
- c. Write a note on Faraday's rotation Isolator. (04 Marks)

Module-3

- 5 a. Explain the construction of micro-strip line. (05 Marks)
- b. Discuss the different types of losses occurs in micro-strip lines. (05 Marks)
- c. A micro-strip line composed of zero – thickness copper conductors on a substrate having $\epsilon_r = 8.4$, $\tan \delta = 0.0005$ and thickness 2.4 mm. If the line width is 1 mm and operated at 10 GHz, calculate :
i) Characteristics impedance Z_0
ii) Attenuation due to conductor loss and dielectric loss. (10 Marks)

OR

- 6 a. Define following parameters with respect to antenna.
i) Radiation pattern
ii) Radiation intensity
iii) Beam Area (Ω_A). (08 Marks)
- b. Explain the radio-communication link and derive Friis transmission formula. (06 Marks)
- c. An antenna has normalized field pattern given by $E_n = \cos^3 \theta$; where θ is polar angle in spherical co-ordinates and it varies from 0 to π . Find the HPBW and directivity. (06 Marks)

Module-4

- 7 a. Derive an expression for radiation resistance of a short-dipole antenna. (08 Marks)
- b. Explain and derive the array of two isotropic point sources of same amplitudes and phase. (08 Marks)
- c. Explain the principle of pattern multiplication. (04 Marks)

OR

- 8 a. Explain with neat diagram linear array of n-isotropic point sources of equal amplitude and spacing. (10 Marks)
- b. Write a note on short dipole antenna. (05 Marks)
- c. Write short notes on Thin linear antenna. (05 Marks)

Module-5

- 9 a. Derive an expression for radiation resistance of a small-loop antenna. (10 Marks)
- b. Explain the rectangular horn antenna and its basic types. (10 Marks)

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- 10 a. Explain the operational modes of a Helicast antenna. (10 Marks)
- b. Explain Yagi-Uda array with the help of neat diagram. (10 Marks)
